

ELEC/PHYS 450/550 — Quantum Computing

Programming Assignment 1: Basic Quantum Circuits and Single-Qubit Operations

Spring 2026 — Dr. Mehmet Cengiz Onbaşlı, Assignment due: March 12, 2026, 23:59

Overview

This assignment covers the foundational topics from Weeks 1-4 of the course: quantum states and superposition, single-qubit Pauli and Hadamard gates, two-qubit CNOT gates, Bell state creation, entanglement measurement, and measurement probability analysis. You will implement all solutions using IBM Qiskit.

Deliverables

- **Completed Jupyter Notebook** (PA1_Quantum_Circuits_Student_ID_number.ipynb) with all code cells filled in and all auto-check cells passing without errors.
Example file name: PA1_Quantum_Circuits_0079989.ipynb
- **Exported PDF** of the completed notebook showing all outputs. Use File → Export Notebook As → PDF in Jupyter or print to PDF.
- Submit both files through KUHUB Learn before the deadline. Do not forget to sign the Honor Code with your name and student ID within the Jupyter notebook.

Questions Summary

#	Topic	Key Concepts	Points
1	Quantum States & Superposition	$ 0\rangle$, $ 1\rangle$, $ +\rangle$, $ -\rangle$; probabilities	20
2	Single-Qubit Gate Properties	Pauli-X/Y/Z, Hadamard matrices; algebraic identities	20
3	Bell State Creation	All 4 Bell states with H + CNOT; circuit drawing	20
4	Entanglement Measure	Von Neumann entropy; reduced density matrix	20
5	Measurement Probability Analysis	Multi-gate circuit; theory vs. simulation	20

Instructions

- Install Qiskit before starting: `pip install qiskit`
- Complete only the cells marked with **### YOUR CODE HERE ###**. Do not modify cells labeled DO NOT MODIFY.
- Each question has an auto-check cell. All assertions must pass before submission.
- Before submission, restart the kernel and run all cells (Kernel → Restart & Run All) to ensure reproducibility.
- Fill in your name and student ID in the first cell of the notebook.

Honor Code

By submitting this assignment you affirm: "I hereby certify that I have completed this assignment on my own without any help from anyone else. I have not used, accessed, received, or distributed any information from/to any other unauthorized source. The effort in this assignment belongs only to me."

Grading

Each question is worth 20 points (100 total). Grading criteria include correctness of code, passing all auto-checks, proper use of Qiskit API, and code clarity. This assignment contributes to the 20% of the 30% homework component of your final grade.